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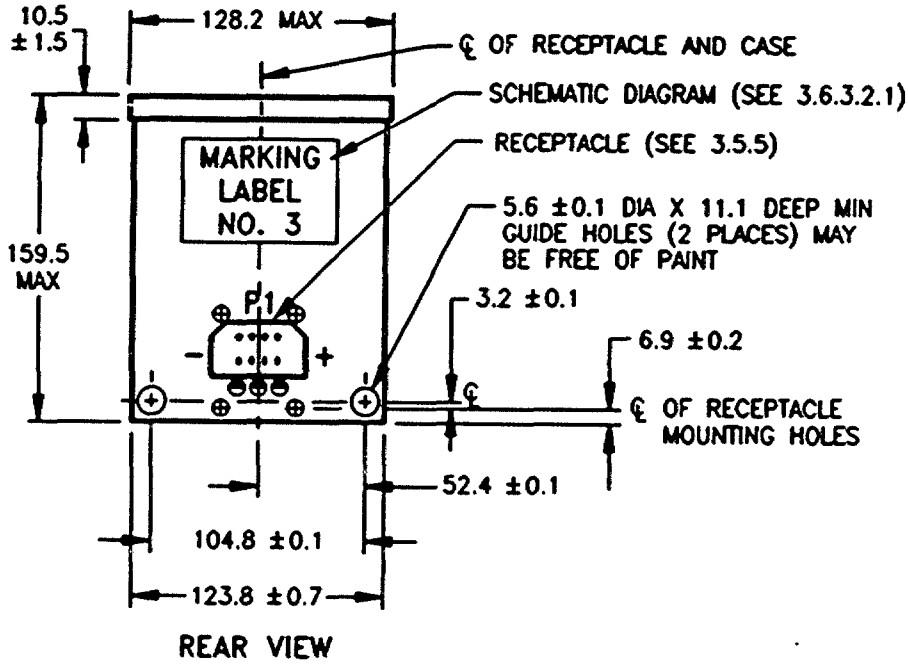
MIL-B-8565/13
21 April 1994

# MILITARY SPECIFICATION SHEET

## BATTERY, STORAGE, AIRCRAFT, MEDIUM-RATE, TYPE 1, MAINTENANCE-FREE, 24-VOLT, 10-AMPERE-HOUR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation:
MIL-B-8565



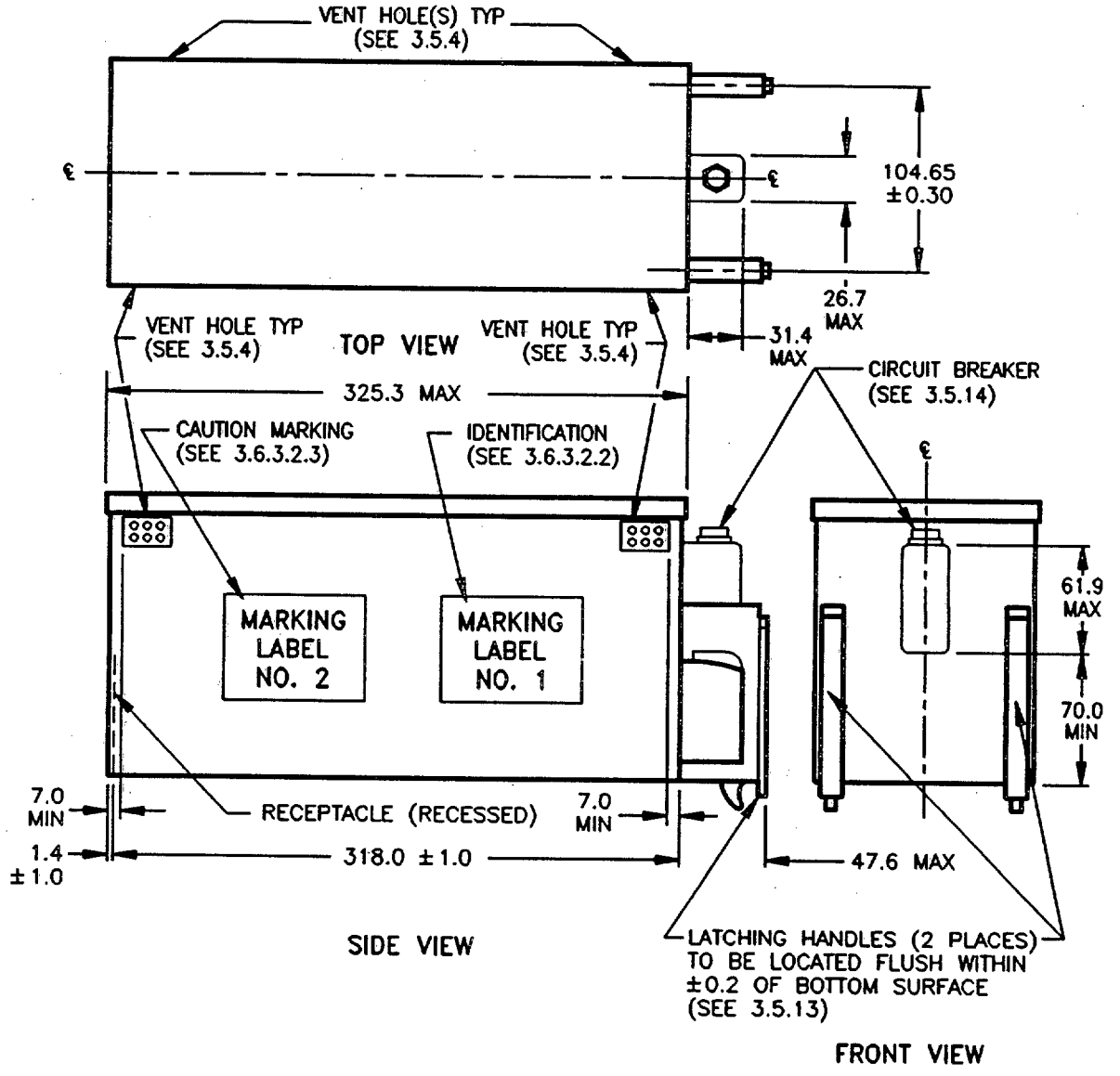

NOTES:
1. All dimensions are in millimeters.
2. Abbreviations used in this figure are diameter (DIA), maximum (MAX), minimum (MIN), and part number (PN).

FIGURE 1. PN D8565/13-1 dimensions.

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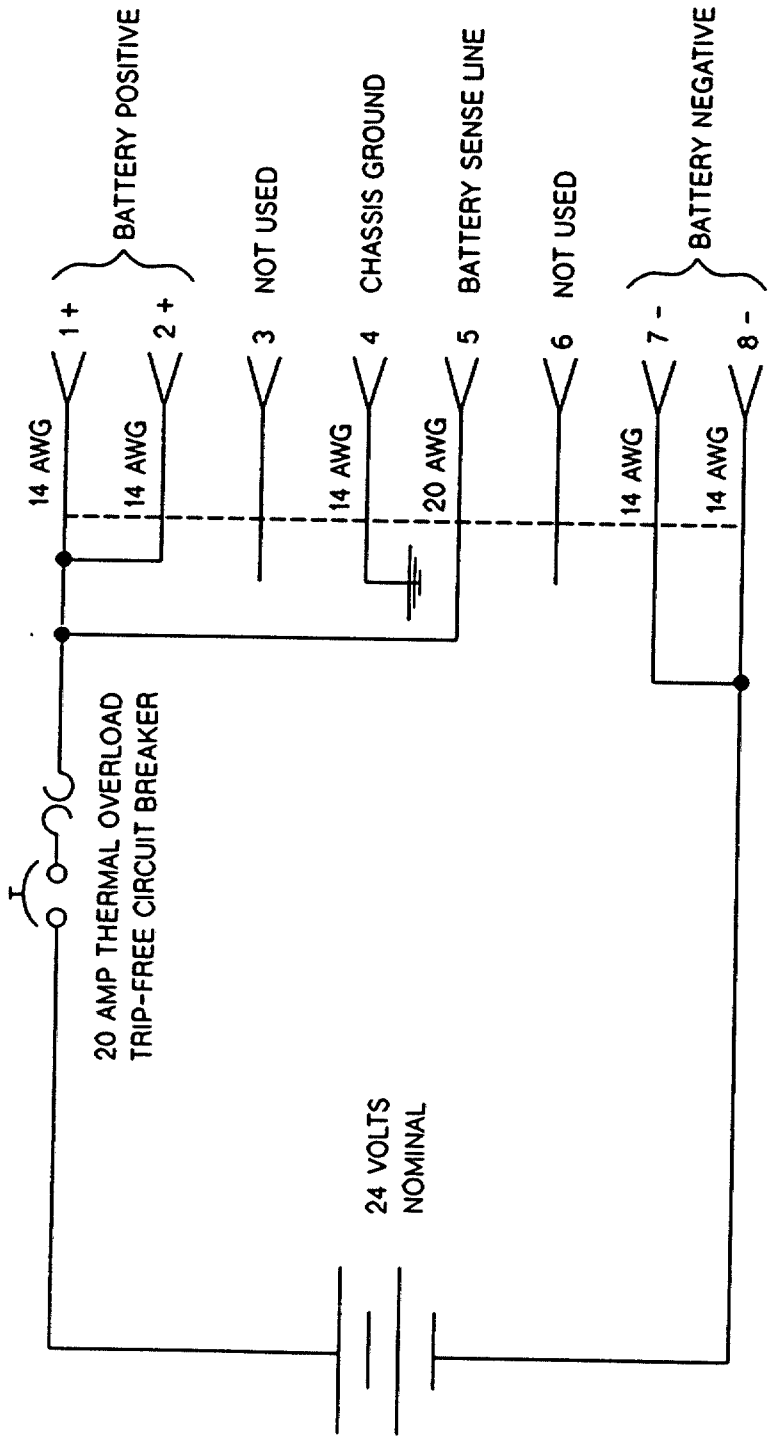

NOTES:

1. All dimensions are in millimeters.
2. Abbreviations used in this figure are maximum (MAX), minimum (MIN), part number (PN) and typical (TYP).

FIGURE 1. PN D8565/13-1 dimensions-Continued.



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NOTES:
1. Abbreviations used in this figure are American wire gauge (AWG) and ampere (AMP).
2. Wire sizes are minimum.

FIGURE 2. Wiring diagram.



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REQUIREMENTS:

1. The dimensions of the battery shall be as shown on Figure 1.

2. The part number of the battery shall be D8565/13-1.

3. The rated capacity of the battery shall be 10 ampere-hours, based on its ability to conform to the life cycle final discharge (10.0 Ah/1 Hr/24°C/18.0 V).

4. The weight of the battery shall be not greater than 14.03 kilograms (31.0 pounds).

5. Battery service life in the aircraft shall be 4 years.

6. Table I lists the quality assurance tests.

7. Delete all requirements for threaded-post connected batteries.

8. The battery shall conform to MIL-B-8565 for type 1, medium rate, except as modified by the following.

2.2 Other publication. Add the following: "ARINC Specification 404".

3.5.2.1 Type 1 battery. Add the following: "The battery container shall meet the requirements of ARINC Specification 404 for a 1/2 ATR short case size, except as modified herein."

3.5.4 Venting. Delete and substitute the following: "The battery case shall contain one or more holes for venting of gases, as depicted on Figure 1. Vents shall be located such that the container cover, battery monoblocs, or cell assemblies do not block the venting holes. If the hole(s) size exceeds 6.5 millimeters, the hole(s) shall be covered by a plastic or wire screen to prevent debris from entering the battery. The cells shall be equipped with a resealable venting mechanism to prevent excessive internal pressure build-up. When the battery is operating within the requirements of this specification, cell venting shall not cause the battery to discharge entrained electrolyte."

3.5.5 Receptacles. Add the following: "The receptacle for the battery's positive and negative connections shall be a Cannon part number DPXB-8-34P-0101-A152 or equivalent. The receptacle shall be located as shown on Figure 1, and wiring connections shall conform to Figure 2."

Add as new requirement:

"3.5.13 Latching handles. The battery shall be equipped with two lever latch handles, Camloc part number 61L2-1-2AA or equivalent. The handles shall be located as shown on figure 1.

"3.5.14 Circuit breaker. The battery shall be equipped with a 20-ampere thermal overload (trip-free) circuit breaker, type MS3320-20. The circuit breaker shall be located as shown on Figure 1, and wiring connections shall conform to Figure 2. The circuit breaker shall be



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mounted in a suitable housing extending from the face of the battery case, provided that the total extension (housing plus circuit breaker) does not protrude above the top surfaces of the battery when the circuit breaker is tripped and provided that the housing does not interfere with the operation of the latches."

3.6.3.2.1 Polarity marking. Delete and substitute the following: "The container shall be conspicuously and durably marked with "+" and "-" in white as shown on Figure 1. The receptacle reference designation "P1" shall be located as shown on Figure 1, with white marking not less than 3 millimeters high. In addition, the schematic diagram of Figure 2 shall be placed on label no. 3 and located as shown on Figure 1."

3.6.3.2.3 Battery caution marking (type 1 battery only). Delete and substitute the following: "Marking for Label 2 shown on Figure 1 shall show the following with lettering height 0.25 inch minimum.

MAINTENANCE-FREE BATTERY

DO NOT REMOVE COVER

PROCESS THIS BATTERY IN ACCORDANCE WITH
APPROVED PROCEDURES ONLY

SEE NAVAIR 17-15BAD-1
OR APPLICABLE SERVICE T.O."

3.6.7 Vent tubes. Delete.

3.6.8 Capacity and electrical performance. Delete TABLE II and substitute the following:

"TABLE II. Capacity performance requirements (type 1 battery).

| Requirement no. | Discharge cutoff voltage and temperature | Rate of discharge factor | Minimum time to cutoff voltage |
|---|---|---|---|
| (1) | 18.0 V at 24 ± 3°C | 1.0C | 60 minutes |
| (2) | 18.0 V at 49 ± 2°C | 1.0C | 66 minutes |
| (3) | 21.0 V at -18 ± 2°C | 400 watts 1/ | 20 minutes |
| (4) | 21.0 V at 5 ± 2°C | 400 watts 1/ | 30 minutes |
| (5) | 21.0 V at -40 ± 2°C | 400 watts | 5 minutes |

1/ During the first minute of discharge, the discharge rate shall be 425 watts. The minimum voltage of 21.0 volts shall apply during the entire discharge period."

3.6.9 Strength of receptacle. Delete.



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3.6.10 Life. Delete and substitute: "Batteries delivered under this specification shall be capable of at least 4 years of service life, demonstrated by successfully completing the cycling test of 4.6.13.1 and the float life test of 4.6.13.4."

Add as a new requirement:

"3.6.10.4 Float life. When tested in accordance with 4.6.13.4, batteries shall provide not less than rated capacity after 50 days of float charging at a temperature of 50°C and a charging rate of 40 milliamperes."

3.6.11 "Evaluation of Equipment" test. Delete and substitute the following: "After all environmental tests, the battery shall be subjected to a 400-watt discharge for 5 minutes. The battery voltage shall be 21.0 volts or greater at the end of the 5 minutes."

3.6.13 Temperature rise and float. Delete and substitute the following: "When tested in accordance with 4.6.21, the battery voltage shall be recorded at intervals not to exceed 5 minutes. The battery voltage, during the 8 hours of constant current charge at 2.0 amperes, shall remain above 29.0 volts once 110 percent of the capacity removed during discharge has been returned. The battery shall meet requirement (1) of table II following the charge period."

4.5.1.3 Constant-potential charging. Delete and substitute the following: "Unless otherwise specified, the battery shall be charged at a constant current rate of 2.00 ± 0.05 amperes until the battery voltage reaches 29.0 ± 0.1 volts. The charging shall then be continued for 2 hours at 29.0 ± 0.1 volts."

4.6.7 Strength of vent tubes. Delete.

4.6.11 Constant voltage discharge (14.0 volts). Delete.

4.6.12 Strength of receptacle. Delete.

4.6.13.1 Cycling test (type 1 battery). Delete "2-hour charge per 4.5.1.3" and substitute "charge in accordance with 4.5.1.3".

Add as new requirement:

"4.6.13.4 Float life. The battery shall be charged at 40 milliamperes at 50 ± 2°C for a period of 50 days. At the end of this time period, the battery shall be stabilized at room temperature and discharged in accordance with 4.6.10. The battery shall comply with the requirements of 3.6.10.4."

4.6.15 Charge and discharge test at low temperature. Delete steps (a) through (d) and substitute the following:

"a. Charge the battery in accordance with 4.5.1.3.



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"b. Place the battery in a test chamber in accordance with 4.5.1.2 at -18°C, then discharge at 400 watts for 15 minutes.

"c. Remove the battery from the chamber and charge for 2.0 hours at 27.0 ± 0.1 volts with the current source limited to 2.0 ± 0.1 amperes.

"d. Immediately following charge, discharge the battery at 400 watts to 21.0 volts. Battery shall meet the requirement (3) of table II."

4.6.16 Discharge while inverted. Delete and substitute the following: "The battery shall be charged as specified in 4.5.1.3. The battery then shall be discharged at 400 watts for 5 minutes. During the first 2.5 minutes of discharge, the battery shall be placed in the inverted position. The battery shall meet the requirements of 3.5.8, and the battery voltage shall remain above 21.0 volts during the entire discharge period."

4.6.20 Special tests. Add the following:

"4.6.20.1 Battery output performance. Battery samples 1, 2 and 3 shall be tested as follows to verify conformance to paragraph 3.6.8.

"a. Charge each battery in accordance with paragraphs 4.5.1.3.

"b. Stabilize the three samples in accordance with 4.5.1.2 at the applicable test temperatures as specified in requirements (3), (4) and (5) of table II.

"c. Discharge sample 1 in accordance with requirement (3) of table II. Discharge sample 2 in accordance with requirement (4) of table II. Discharge sample 3 in accordance with requirement (5) of table II.

"d. Each sample shall meet the minimum cutoff voltage of the applicable requirements of table II."

4.6.21 Temperature rise and float test. Delete and substitute the following: "The battery shall be charged in accordance with 4.5.1.3 and then placed in a temperature chamber at 49 ± 2°C for 12 hours. At this temperature, the battery shall be discharged at 400 watts for 30 minutes. Immediately following this discharge, with the battery still in the chamber at 49 ± 2°C, a constant current charge of 2.0 ± 0.1 amperes shall be applied for 8.0 ± 0.1 hours. The battery shall then be stabilized at 24 ± 5°C and discharged in accordance with 4.6.10. The battery shall meet the requirements of 3.6.12 and 3.6.13."



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TABLE I. Inspection of batteries and order of test.

| Order of test | Description of inspection | Requirements paragraph | Inspection method paragraph | Qualification inspection | | | | Quality conformance inspection | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Qualification sample number | | | | Group A tests | Group B tests | |
| | | | | 1 | 2 | 3 | 4 | | 1 | 2 |
| 1 | Visual & mechanical examination | 3.5, 3.6.1, 3.6.3 & 3.6.22 | 4.6.3 | X | X | X | X | X | X | X |
| 2 | Dimensions & weight | 3.6.5 | 4.6.4 | X | X | X | | X | X | X |
| 3 | Dielectric strength | 3.6.4 | 4.6.5 | X | X | X | X | X | X | X |
| 4 | Color & marking | 3.6.3 | 4.6.8 | X | X | X | | X | X | X |
| 5 | Conditioning charge | 4.5.1.4 | 4.6.9 | X | X | X | X | | X | X |
| 6 | Capacity discharge | 3.6.8 | 4.6.10 | X | X | X | X | X | X | X |
| 7 | Battery output performance | 3.6.8 | 4.6.20.1 | X | X | X | | | X | X |
| 8 | Charge & discharge test at low temperature | 3.6.8 | 4.6.15 | | | X | | | X | |
| 9 | Life cycling | 3.6.10.1 | 4.6.13.1 | X | | | | | X | |
| 10 | Float life | 3.6.10.4 | 4.6.13.4 | | X | | | | | |
| 11 | Capacity discharge at 49°C | 3.6.8 | 4.6.14 | | X | | | | | X |
| 12 | Discharge while inverted | 3.6.8 & 3.6.11 | 4.6.16 | | | X | | | X | |
| 13 | Altitude | 3.6.11 & 3.6.12 | 4.6.17 | | | X | | | | X |
| 14 | Mechanical shock | 3.6.11 & 3.6.12 | 4.6.18 | X | | | | | | |
| 15 | Temperature shock | 3.6.11 & 3.6.12 | 4.6.19 | X | | | | | X | |
| 16 | Temperature rise & float | 3.6.11, 3.6.12 & 3.6.13 | 4.6.21 | | | X | | | | X |
| 17 | Battery gas emission test | 3.6.14 | 4.6.22 | | | X | | | | |
| 18 | Vibration | 3.6.11, 3.6.12 & 3.6.16 | 4.6.23 | | | | X | | | |
| 19 | Humidity | 3.6.11, 3.6.12 & 3.6.17 | 4.6.24 | | X | | | | | X |
| 20 | Salt fog | 3.6.11, 3.6.12 & 3.6.18 | 4.6.25 | X | | | | | X | |
| 21 | Ground storage | 3.6.15 | 4.6.26 | | | | X | | | |
| 22 | Physical integrity at high temperature | 3.6.12 | 4.6.27 | | X | | | | | |
| 23 | Final examination | 3.6.4 | 4.6.30 | X | X | X | X | | X | X |





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